

Chapter 5 The Indefinite Integral

Exercises 5.1 Concept and Property of the Indefinite Integral

1. Completion

$$\frac{d}{dx} \int f(x) dx = f(x)$$

Assume $f(x)$ is derivable, then $d \int f(x) dx = \underline{f(x)} dx$; $\int d f(x) = \underline{f(x)} + C$

$$\frac{d}{dx} \int f(x) dx = \underline{f(x)}; \int f'(x) dx = \underline{f(x)} + C$$

2. The curve with slope $2x$ passes through the point (2,5), find the curve.

$$y = 2x, \quad y = x^2 + C, \quad y(2) = 5 \Rightarrow C = 1,$$

$$\therefore y = x^2 + 1$$

3. Calculate the following indefinite integrals with the Table of indefinite integral Formulas.

$$(1) \int \frac{(1+\sqrt{x})^2}{x} dx:$$

$$\begin{aligned} &= \int \left(\frac{1}{x} + \frac{2}{\sqrt{x}} + 1 \right) dx \\ &= \ln|x| + 4\sqrt{x} + x + C \end{aligned}$$

$$(2) \int (e^x - 3 \cos x) dx:$$

$$\begin{aligned} &= \int \left(\frac{1}{x} + \frac{2}{\sqrt{x}} + 1 \right) dx \\ &= e^x - 3 \sin x + C \end{aligned}$$

$$(3) \int \tan^2 x dx:$$

$$\begin{aligned} &= \int (\sec^2 x - 1) dx \\ &= \tan x - x + C \end{aligned}$$

$$(4) \int \sin^2 \frac{x}{2} dx:$$

$$\begin{aligned} &= \int \frac{1 - \cos x}{2} dx = \frac{x}{2} - \frac{1}{2} \sin x + C \end{aligned}$$

$$(5) \int \frac{\cos 2x}{\cos x - \sin x} dx:$$

$$\begin{aligned} &= \int \frac{\cos^2 x - \sin^2 x}{\cos x - \sin x} dx \\ &= \int (\cos x + \sin x) dx = \sin x - \cos x + C \end{aligned}$$

$$(6) \int \frac{\sqrt{x^4 + x^2 + 2}}{x^3} dx.$$

$$\begin{aligned} &= \int \frac{x^2 + x^{-2}}{x^3} dx = \int \left(\frac{1}{x} + x^{-5} \right) dx \\ &= \ln|x| - \frac{1}{4} x^{-4} + C \end{aligned}$$

1. 求导

4. (1) If $F(x)$ is an antiderivative of $\frac{\ln x}{x}$, find $dF(\sin x)$:

$$F'(x) = \frac{\ln x}{x}$$

$$dF(\sin x) = F'(\sin x) \cos x dx$$

$$= \frac{\ln(\sin x)}{\sin x} \cos x dx = \cot x \cdot \ln(\sin x) dx$$

反导 (即积分)

- (2) If $\frac{\sin x}{1+x \sin x}$ is an antiderivative of $f(x)$, find $\int f(x)f'(x)dx$.

$$\left(\frac{\sin x}{1+x \sin x} \right)' = f(x)$$

$$f(x) = \left(\frac{\sin x}{1+x \sin x} \right)' = \frac{\cos x \cdot (1+x \sin x) - \sin x \cdot (\sin x + x \cos x)}{(1+x \sin x)^2}$$

$$= \frac{\cos x - \sin^2 x}{(1+x \sin x)^2}$$

$$\int f(x)f'(x)dx$$

$$= \int f(x) df(x) = \frac{1}{2} f^2(x) + C = \frac{(\cos x - \sin^2 x)^2}{2(1+x \sin x)^2} + C$$

2

Exercises 5.2 Integration by Substitutions

Calculate the following indefinite integrals with the Substitution Rule.

$$(1) \int \frac{1}{e^x + e^{-x}} dx:$$

$$= \int \frac{e^x}{e^{2x} + 1} dx \quad u = e^x$$

$$= \int \frac{1}{(1+u^2)} du$$

$$= \arctan u + C = \arctan e^x + C$$

$$(3) \int \frac{x}{\sqrt{1-x^2}} dx:$$

$$= -\frac{1}{2} \int (1-x^2)^{-\frac{1}{2}} d(1-x^2)$$

$$= -\sqrt{1-x^2} + C$$

$$(4) \int \frac{\cot x}{\ln \sin x} dx:$$

$$= \int \frac{\cos x}{\sin x \ln \sin x} dx$$

$$= \int \frac{d(\ln \sin x)}{\ln \sin x} = \ln |\ln \sin x| + C$$

$$(5) \int \frac{\sin^2 x \cos x}{1 + \sin^2 x} dx:$$

$$= \int \frac{\sin^2 x + 1 - 1}{1 + \sin^2 x} d(\sin x)$$

$$= \int (1 - \frac{1}{1 + \sin^2 x}) d(\sin x)$$

$$= \sin x - \arctan(\sin x) + C$$

3

1901815.2.12: A2000

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln(\sqrt{x^2+a^2}+x) + C$$

$$(7) \int \frac{1}{\sqrt{x(1+x)}} dx :$$

$$= 2 \int \frac{d(\sqrt{x})}{1+(\sqrt{x})^2}$$

$$= 2 \arctan \sqrt{x} + C$$

$$(8) \int \frac{1}{x\sqrt{x^2+1}} dx = \int \frac{dx}{x|x|\sqrt{1+(\frac{1}{x})^2}}$$

$$= \int \frac{dx}{\sqrt{1+(\frac{1}{x})^2}} = -\operatorname{sgn} x \int \frac{d(\frac{1}{x})}{\sqrt{1+(\frac{1}{x})^2}}$$

$$= -\operatorname{sgn} x \cdot \ln\left(\frac{1}{x} + \sqrt{1+\left(\frac{1}{x}\right)^2}\right) + C$$

$$(9) \int \frac{1}{\sqrt{3+2x-x^2}} dx :$$

$$= \int \frac{d(x-1)}{\sqrt{4-(x-1)^2}}$$

$$= \arcsin \frac{x-1}{2} + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$$

$$(11) \int \frac{1}{x\sqrt{x^2-4}} dx :$$

$$= \int \frac{\frac{x^2-4}{x} \cdot \frac{1}{x} dx}{\sqrt{\frac{x^2-4}{x} \cdot \frac{1}{x}}} = \int \frac{x^2-4}{x\sqrt{x^2-4}} dx$$

$$= \int \frac{\sqrt{x^2-4} - 4 \operatorname{sgn} x}{\sqrt{x^2-4}} dx$$

$$= \int \frac{1}{\sqrt{x^2-4}} dx - 4 \operatorname{sgn} x \int \frac{1}{\sqrt{x^2-4}} dx$$

$$= \frac{1}{2} \ln \left| \frac{x-2}{x+2} \right| - 4 \operatorname{sgn} x \cdot \frac{1}{2} \ln \left| \frac{x-2}{x+2} \right| + C$$

$$(10) \int \frac{x^2}{\sqrt{9-x^2}} dx : \quad x=3\sin t, \quad 1 \leq t < \frac{\pi}{2},$$

$$= \int \frac{9\sin^2 t}{3\cos t} \cdot 3\cos t dt = 9 \int \sin^2 t dt = \frac{9}{2} t - \frac{9}{4} \sin 2t + C = \frac{9}{2} t - \frac{9}{2} \sin t \cos t + C$$

$$= \frac{9}{2} \arcsin \frac{x}{3} - \frac{1}{2} x \sqrt{9-x^2} + C$$

$$(12) \int \frac{1}{x^2\sqrt{x^2+9}} dx :$$

$$= \operatorname{sgn} x \int \frac{x^{-3} dx}{\sqrt{1+9x^{-2}}}$$

$$= -\frac{1}{8} \operatorname{sgn} x \int \frac{d(1+9x^{-2})}{\sqrt{1+9x^{-2}}}$$

$$= -\frac{1}{9} \operatorname{sgn} x \sqrt{1+9x^{-2}} + C$$

$$= -\frac{\sqrt{x^2+9}}{9x} + C$$

Exercises 5.3 Integration by Parts

Calculate the following indefinite integrals with The Rule for Integration by Parts.

$$(1) \int (\ln x)^2 dx :$$

$$= x(\ln x)^2 - \int x \cdot 2 \ln x \cdot \frac{1}{x} dx$$

$$= x(\ln x)^2 - 2 \int \ln x dx = x(\ln x)^2 - 2(x \ln x - \int x \cdot \frac{1}{x} dx)$$

$$= x(\ln x)^2 - 2x \ln x + 2x + C$$

$$(2) \int x^2 e^{-x^2} dx = -\frac{1}{2} \int x^2 e^{-x^2} dx$$

$$= -\frac{1}{2} \int x^2 d(e^{-x^2}) = -\frac{1}{2} (x^2 e^{-x^2} - \int e^{-x^2} d(x^2))$$

$$= -\frac{1}{2} x^2 e^{-x^2} + \frac{1}{2} \int e^{-x^2} dx = -\frac{1}{2} x^2 e^{-x^2} + \frac{1}{2} \sqrt{\pi} \operatorname{erf}(x) + C$$

$$= -\frac{1}{2} x^2 e^{-x^2} + \frac{1}{2} \sqrt{\pi} \operatorname{erf}(x) + C$$

$$(4) \int x^2 \ln x dx = \frac{1}{3} \int \ln x d(x^3)$$

$$= \frac{1}{3} x^3 \ln x - \frac{1}{3} \int x^3 \cdot \frac{1}{x} dx = \frac{1}{3} x^3 \ln x - \frac{1}{9} x^3 + C$$

$$= \frac{1}{3} x^3 \ln x - \frac{1}{9} x^3 + C$$

$$= e^x \sin x - e^x \cos x + \int e^x d \cos x = e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

$\int \frac{dx}{(t^2+1)^2} = \int \frac{1}{(\cos^2 x)^2} \frac{dx}{\cos x} = \int \frac{\sec^3 x}{\cos x} dx = \int \sec^4 x dx = \int \sec^2 x \cdot \sec^2 x dx$
 $= \int \sec^2 x \cdot (1 + \tan^2 x) dx = \int \sec^2 x dx + \int \sec^2 x \tan^2 x dx$
 $= \tan x + \int \sec^2 x \tan^2 x dx$
 $\text{Let } u = \tan x, du = \sec^2 x dx$
 $= \tan x + \int u^2 du = \tan x + \frac{1}{3} u^3 + C = \tan x + \frac{1}{3} \tan^3 x + C$

Exercises 5.4 Integration of Special Functions

1. Find the integration of rational functions.

(1) $\int \frac{x}{1+x} dx$

$= \int \left(\frac{x^2 - x + 1}{x+1} - \frac{1}{x+1} \right) dx$
 $= \int \left(\frac{x^2 - x + 1}{x+1} \right) dx - \int \frac{1}{x+1} dx$
 $= \frac{1}{3} x^3 - \frac{1}{2} x^2 + x - \ln|x+1| + C$
 $= \frac{1}{3} x^3 - \frac{1}{2} x^2 + x - \ln|x+1| + C$

(2) $\int \frac{x^2 + x - 8}{x^3 - x} dx$

(3) $\int \frac{x^3 + 1}{x^3 - x^2} dx$

$= \int \left(\frac{x^3 + 1}{x^2(x-1)} \right) dx = \int \left(\frac{1}{x-1} + \frac{1}{x} + \frac{1}{x^2} \right) dx$
 $= \ln|x-1| + \ln|x| - \frac{1}{x} + C$

(4) $\int \frac{x^2}{1-x^2} dx = \frac{1}{2} \int \left(\frac{1}{1-x^2} - \frac{1}{1+x^2} \right) dx$

$= \frac{1}{2} \left(\ln \left| \frac{1+x}{1-x} \right| - \arctan x \right) + C$
 $= \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| - \frac{1}{2} \arctan x + C$

$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C, a \neq 0$
 $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan \frac{x}{a} + C, a \neq 0$

(1) $\int \frac{1}{x\sqrt{x+1}} dx$

$\text{Let } u = \sqrt{x+1}, du = \frac{1}{2\sqrt{x+1}} dx$
 $= \int \frac{2 du}{u^2} = -\frac{2}{u} + C = -\frac{2}{\sqrt{x+1}} + C$

(2) $\int \frac{\sqrt{x}}{(1+x)^2} dx$

$\text{Let } u = \sqrt{x}, x = u^2, dx = 2u du$
 $= \int \frac{u}{(1+u^2)^2} \cdot 2u du = 2 \int \frac{u^2}{(1+u^2)^2} du$
 $= 2 \int \frac{(1+u^2) - 1}{(1+u^2)^2} du = 2 \int \left(\frac{1}{1+u^2} - \frac{1}{(1+u^2)^2} \right) du$
 $= 2 \left(\arctan u - \frac{u}{2(1+u^2)} \right) + C$
 $= 2 \arctan \sqrt{x} - \frac{\sqrt{x}}{1+x} + C$

$= \ln \left| \frac{x+1}{x-1} \right| - 2 \arctan \sqrt{\frac{x+1}{x-1}} + C$
 $= \ln \left| \frac{x+1}{x-1} \right| - 2 \arctan \sqrt{\frac{x+1}{x-1}} + C$

$\sin t = \tan t \cdot \cos t$

$\int \frac{dx}{\cos^2 t} = \int \sec^2 t dt = \tan t + C$
 $\text{Let } u = \tan t, du = \sec^2 t dt$
 $= \int \frac{1}{u^2} du = -\frac{1}{u} + C = -\frac{1}{\tan t} + C = -\cot t + C$

(3) $\int \frac{1}{x^4 + 1 + x^2} dx$

$= \int \frac{1}{x^4 + x^2 + 1} dx = \int \frac{1}{(x^2 + \frac{1}{2})^2 + \frac{3}{4}} dx$
 $= \int \frac{1}{(x^2 + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} dx$
 $= \frac{2}{\sqrt{3}} \int \frac{1}{(x^2 + \frac{1}{2})^2 + 1} dx$
 $= \frac{2}{\sqrt{3}} \left(\frac{1}{2} \arctan \frac{x}{1} + \frac{x}{2(1+x^2)} \right) + C$
 $= \frac{1}{\sqrt{3}} \arctan x + \frac{x}{1+x^2} + C$

3. Find the integration of trigonometric rational functions.

(1) $\int \frac{1}{1 + \sin x + \cos x} dx$

$\text{Let } t = \tan \frac{x}{2}, \sin x = \frac{2t}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2}, dx = \frac{2}{1+t^2} dt$
 $= \int \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$
 $= \int \frac{2}{1+t^2 + 2t + 1 - t^2} dt = \int \frac{2}{(t+1)^2} dt$
 $= -\frac{2}{t+1} + C = -\frac{2}{1 + \tan \frac{x}{2}} + C$

(2) $\int \frac{1}{\sin x + \tan x} dx$

$\text{Let } t = \tan \frac{x}{2}, \sin x = \frac{2t}{1+t^2}, \tan x = \frac{t}{1-t^2}, dx = \frac{2}{1+t^2} dt$
 $= \int \frac{1}{\frac{2t}{1+t^2} + \frac{t}{1-t^2}} \cdot \frac{2}{1+t^2} dt$
 $= \int \frac{2}{2t(1-t^2) + t(1+t^2)} dt = \int \frac{2}{t(1-t^2 + 1+t^2)} dt$
 $= \int \frac{2}{t(2)} dt = \int \frac{1}{t} dt = \ln|t| + C = \ln \left| \tan \frac{x}{2} \right| + C$

(3) $\int \frac{1}{5 + 4 \sin x} dx$

$\text{Let } t = \tan \frac{x}{2}, \sin x = \frac{2t}{1+t^2}, dx = \frac{2}{1+t^2} dt$
 $= \int \frac{1}{5 + \frac{8t}{1+t^2}} \cdot \frac{2}{1+t^2} dt = \int \frac{2}{5(1+t^2) + 8t} dt$
 $= \int \frac{2}{5t^2 + 8t + 5} dt$
 $= \frac{2}{5} \int \frac{1}{t^2 + \frac{8}{5}t + 1} dt$
 $= \frac{2}{5} \int \frac{1}{(t + \frac{4}{5})^2 + \frac{9}{5}} dt$
 $= \frac{2}{5} \cdot \frac{1}{\frac{3}{\sqrt{5}}} \arctan \frac{t + \frac{4}{5}}{\frac{3}{\sqrt{5}}} + C$
 $= \frac{2\sqrt{5}}{15} \arctan \left(\frac{\sqrt{5}}{3} \tan \frac{x}{2} + \frac{4\sqrt{5}}{3} \right) + C$

(4) $\int \frac{1}{\sin^2 x \cos x} dx$

$\text{Let } u = \sin x, du = \cos x dx$
 $= \int \frac{1}{u^2 \cdot \frac{1}{u}} du = \int \frac{1}{u} du = \ln|u| + C = \ln|\sin x| + C$
 $= \ln|\sin x| + C$

$$(\tan x)' = \sec^2 x = \frac{1}{\cos^2 x}$$

$$(5) \int \frac{1}{a^2 \cos^2 x + b^2 \sin^2 x} dx, (a > 0, b > 0):$$

$$\int \frac{1}{a^2 + b^2 \tan^2 x} d(\tan x) = \frac{1}{ab} \int \frac{d(\frac{b}{a} \tan x)}{1 + (\frac{b}{a} \tan x)^2}$$

$$= \frac{1}{ab} \arctan\left(\frac{b}{a} \tan x\right) + C$$

$$(6) \int \frac{\sin^3 x}{2 + \cos x} dx = \int \frac{-\sin^2 x}{2 + \cos x} d(\cos x)$$

$$= \int \frac{\cos^2 x - 1}{\cos x + 2} d(\cos x) = \int \frac{(\cos^2 x - 4) + 3}{\cos x + 2} d(\cos x)$$

$$= \int (\cos x - 2) d(\cos x) + 3 \int \frac{1}{2 + \cos x} d(2 + \cos x)$$

$$= \frac{1}{2} \cos^2 x - 2 \cos x + 3 \ln(2 + \cos x) + C$$

$$F(x) = \int_0^x f(t) dt - \tilde{F}(0) = \int_0^x f(t) dt - \int_0^x f(u) du = \int_0^x f(u) du - \tilde{F}(0) = \tilde{F}(x)$$

Test 5

I. Choices

$$f'(x) = \sin x$$

1. Let $\sin x$ be the derivative of $f(x)$, then the antiderivative of $f(x)$ may be

$$\sin x \int_0^x \sin x dx = (-\sin x)' = -\cos x$$

$$\Rightarrow f'(x) = \sin x \Rightarrow f(x) = -\cos x$$

2. Assume $f(x)$ is an odd function, $F(x)$ is an antiderivative of $f(x)$, then $F(x)$ is

$$\text{neither odd nor even function}$$

(A) an odd function.

(C) neither odd nor even function.

3. Assume $f(x) \geq 0$, $F(x)$ is an antiderivative of $f(x)$, then $F(x)$ is

$$F(x) = \int_0^x f(t) dt - \tilde{F}(0) \uparrow (\because f(t) \geq 0)$$

(A) increasing. (B) decreasing. (C) strictly decreasing. (D) strictly increasing.

4. $F(x)$ is an antiderivative of $f(x)$, then $\int \sin x (\cos x) dx = -\int dF(\cos x)$

$$(D) = -F(\cos x) + C$$

$$(A) F(\sin x) + C.$$

$$(C) F(\cos x) + C.$$

$$5. \int |x| dx = \begin{cases} x > 0, \int x dx = \frac{1}{2} x^2 + C \\ x < 0, \int (-x) dx = -\frac{1}{2} x^2 + C \end{cases}$$

$$(A) |x| + C.$$

$$(C) \frac{1}{2} x^2 + C.$$

$$(D) \frac{1}{2} x|x| + C.$$

II. Calculation problems.

$$1. \text{ Evaluate } \int \frac{1}{1 + \sin x} dx = \int \left(\frac{1}{1 + \sin x} \cdot \frac{1 - \sin x}{1 - \sin x} \right) dx = \int \frac{1 - \sin x}{1 - \sin^2 x} dx$$

$$= \int \frac{1 - \sin x}{\cos^2 x} dx = \int \sec^2 x dx - \int \tan x \cdot \sec x dx$$

$$= \tan x - \sec x + C$$

$$\because (\tan x)' = \sec^2 x, (\sec x)' = \tan x \cdot \sec x$$

$$\frac{d}{dx} \left(\frac{\cos x}{x^2} \right) = -\frac{\sin x \cdot x - \cos x \cdot 2x}{x^4} = -\frac{\sin x}{x^3} + \frac{2 \cos x}{x^3}$$

2. Evaluate $\int \frac{1}{\sqrt{(x-a)(b-x)}} dx$ ($a < x < b$).

$$\begin{aligned} &= 2 \int \frac{1}{\sqrt{b-a}} \frac{1}{\sqrt{1 - \frac{x-a}{b-a}}} dx \\ &= 2 \int \frac{1}{\sqrt{b-a}} \frac{1}{\sqrt{1 - u}} du \quad \left(u = \frac{x-a}{b-a} \right) \\ &= 2 \int \frac{1}{\sqrt{b-a}} \frac{1}{\sqrt{1-u}} du = 2 \arcsin \sqrt{\frac{x-a}{b-a}} + C \end{aligned}$$

3. Evaluate $\int \frac{x^{1/4}}{(x^2+1)^4} dx$.

$$\begin{aligned} &= -\frac{1}{5} \int (x^2+1)^{-4} d(x^2+1) = -\frac{1}{15} (x^2+1)^{-3} + C \\ &= -\frac{1}{15(x^2+1)^3} + C \end{aligned}$$

4. Assume $f(x)$ is derivable, and $\int x^2 f(x) dx = x^2 \cos x - 4x \sin x - 6 \cos x + C$, find

$$\begin{aligned} f(x) &= \frac{d}{dx} (x^2 \cos x - 4x \sin x - 6 \cos x) \\ &= 2x \cos x - x^2 \sin x - 4 \sin x + 4x \cos x + 6 \sin x \\ &= 6 \sin x + 2x \cos x - 4 \sin x \\ &= 2 \sin x + 2x \cos x \end{aligned}$$

5. Find the coefficients A, B such that

$$\frac{1}{(a+b \cos x)^2} = \frac{A \sin x}{a+b \cos x} + \frac{B}{a+b \cos x}$$

$$\begin{aligned} 1 &= A a \cos x + A b + B a + B b \cos x \\ \Rightarrow \begin{cases} A a + B a = 1 \\ A b + B b = 0 \end{cases} &\Rightarrow \begin{cases} A = \frac{1}{b-a^2} \\ B = -\frac{a}{a^2-b^2} \end{cases} \end{aligned}$$

$$a_1 + a_2 + \dots + a_n = \frac{a_1 (1 - q^n)}{1 - q} = \frac{a_1}{1 - q} (1 - q^n)$$

Chapter 6 The Definite Integral

Exercises 6.1 Concept of The Definite Integral

1. Use the definition of the definite integral to calculate the following integrals.

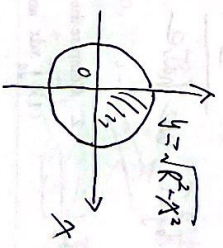
$$\begin{aligned} (1) \int_1^2 x^2 dx &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(1 + \frac{k}{n} \right)^2 \cdot \frac{1}{n} \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{1}{n} + \frac{2k}{n^2} + \frac{k^2}{n^3} \right) \\ &= \lim_{n \rightarrow \infty} \left(1 + \frac{2}{n} \cdot \frac{n(n+1)}{2n} + \frac{1}{6} \frac{n(n+1)(2n+1)}{6n^3} \right) \\ &= 1 + 1 + \frac{1}{3} = \frac{7}{3} \end{aligned}$$

2. State the following equations with the geometric meaning of definite integral.

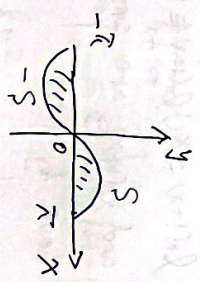
(1) $\int_0^R \sqrt{R^2 - x^2} dx = \frac{1}{4} \pi R^2$

(2) $\int_{-\pi}^{\pi} \sin x dx = 0$

(1) $\int_0^R \sqrt{R^2 - x^2} dx$ is $\frac{1}{4}$ of the area of the circle $x^2 + y^2 = R^2$.



(2) $\int_{-\pi}^{\pi} \sin x dx$ represents the algebraic sum of two blocks of area in the graph.



thus $\int_{-\pi}^{\pi} \sin x dx = 0$

Exercises 6.2 Properties of the Definite Integral

and the Mean-value Theorem

1. Assume $f(x)$ and $g(x)$ are continuous on the interval $[a, b]$, prove that:

Give (1) If $f(x) \leq 0, x \in [a, b]$, and $\int_a^b f(x) dx = 0$, then $f(x) = 0, x \in [a, b]$.

Proof. If there exists $x_0 \in [a, b], f(x_0) < 0$. then for $\varepsilon_0 = -\frac{1}{2}f(x_0) > 0, \exists \delta > 0$, such that $(f(x) - f(x_0)) < \varepsilon_0, |x - x_0| < \delta$.

then $f(x) \leq \frac{1}{2}f(x_0), x \in [x_0 - \delta, x_0 + \delta] \cap [a, b]$, $\Rightarrow f(x) - f(x_0) \leq -\frac{1}{2}f(x_0)$, $\Rightarrow f(x) \leq \frac{1}{2}f(x_0)$.

thus $\int_a^b f(x) dx = \int_{[a, b] \setminus [x_0 - \delta, x_0 + \delta]} f(x) dx + \int_{[x_0 - \delta, x_0 + \delta] \cap [a, b]} f(x) dx \leq 0 + \int_{[x_0 - \delta, x_0 + \delta] \cap [a, b]} \frac{1}{2}f(x_0) dx < 0$, contradiction.

(2) If $f(x) \leq g(x), x \in [a, b]$ and $\int_a^b f(x) dx = \int_a^b g(x) dx$, then $f(x) = g(x)$.

$$\int_a^b f(x) dx < \int_a^b g(x) dx, x \in [a, b].$$

Proof. $f(x) \leq g(x), x \in [a, b], \Rightarrow \int_a^b f(x) dx \leq \int_a^b g(x) dx$.

disproof. If $\int_a^b f(x) dx = \int_a^b g(x) dx$, then $\int_a^b (f(x) - g(x)) dx = 0$.

by (1), we have $f(x) - g(x) \equiv 0$, i.e. $f(x) \equiv g(x), x \in [a, b]$, contradiction.

2. Compare the values of the following integrals according to the properties of definite integrals.

$$(1) \int_0^1 x dx \text{ and } \int_0^1 \ln(1+x) dx;$$

$$(2) \int_0^1 e^x dx \text{ and } \int_0^1 (1+x) dx.$$

$\because \ln(1+x) \leq x, \ln(1+x) \neq x, x \in [0, 1]$

$e^x \geq (1+x), e^x \neq (1+x), x \in [0, 1]$

x and $\ln(1+x)$ are continuous on $[0, 1]$,

e^x and $(1+x)$ are continuous on $[0, 1]$,

We have by (1) (2)

$$\int_0^1 \ln(1+x) dx < \int_0^1 x dx$$

12

We have by (1) (2)

$$\int_0^1 \ln(1+x) dx < \int_0^1 e^x dx$$

$$f(x) = \frac{1}{1+0.5 \cos x} \text{ in } [0, 2\pi] \text{ is } \int_0^{2\pi} f(x) dx$$

$$f(x) = \frac{1}{\sqrt{1+x}} \text{ in } [0, 1] \text{ is } \int_0^1 f(x) dx$$

3. Use the mean value theorem for integrals to estimate the following integrals.

$$(1) \int_0^{2\pi} \frac{1}{1+0.5 \cos x} dx;$$

$$(2) \int_0^1 \frac{x^9}{\sqrt{1+x}} dx;$$

By mean value theorem of integrals

$$I = \int_0^{2\pi} \frac{dx}{1+0.5 \cos x} = \frac{2\pi}{3 \in [0, 2\pi]}, \quad I = \int_0^1 \frac{x^9}{\sqrt{1+x}} dx = \frac{1}{10 \sqrt{1+\xi}}, \quad \xi \in [0, 1]$$

$$\therefore \frac{2\pi}{1+0.5} \leq I \leq \frac{2\pi}{1-0.5}$$

$$\text{i.e. } \frac{4\pi}{3} \leq I \leq 4\pi.$$

$$f(x) = \arctan x \text{ in } [\frac{1}{\sqrt{3}}, \sqrt{3}] \text{ is } \int_{1/\sqrt{3}}^{\sqrt{3}} f(x) dx$$

$$(3) \int_{1/\sqrt{3}}^{\sqrt{3}} \arctan x dx; \quad (4) \int_0^{100} \frac{e^{-x}}{x+100} dx \approx I = \frac{1}{100+3} \int_0^{100} e^{-x} dx$$

$$\text{By mean value theorem of integrals} \quad I = \int_{1/\sqrt{3}}^{\sqrt{3}} \arctan x dx = \frac{1-e^{-100}}{100+3}, \quad \xi \in [0, 100]$$

$$I = \int_{1/\sqrt{3}}^{\sqrt{3}} \arctan x dx = \arctan 3 - \arctan \frac{1}{\sqrt{3}}$$

$$= \frac{4}{3} \arctan 3, \quad \xi \in [\frac{1}{\sqrt{3}}, \sqrt{3}], \quad f(x) = \frac{1}{x+100} \text{ in } [0, 100] \text{ is } \int_0^{100} \frac{1}{x+100} dx$$

$$= \frac{4}{3} \arctan 3, \quad \xi \in [\frac{1}{\sqrt{3}}, \sqrt{3}], \quad \tan \frac{\pi}{6} = \tan 30^\circ = \frac{1}{\sqrt{3}}, \quad \tan \frac{\pi}{3} = \tan 60^\circ = \sqrt{3}$$

$$\therefore \frac{\pi}{6} \leq \arctan 3 \leq \frac{\pi}{3}, \quad I = e^{-3} \int_0^{100} \frac{1}{x+100} dx = e^{-3} \ln 2, \quad \xi \in [0, 100]$$

$$\frac{2\pi}{9} \leq \frac{4}{3} \arctan 3 \leq \frac{4\pi}{9}, \quad \text{i.e. } \frac{2\pi}{9} \leq I \leq \frac{4\pi}{9}$$

$$f(x) = \frac{e^{-x}}{x+100} \text{ in } [0, 100] \text{ is } \int_0^{100} f(x) dx = e^{-100} \ln 2, \quad \xi \in [0, 100]$$

$$t \rightarrow 0 \text{ or } \infty, \quad \arctan t^2 \sim \frac{t^2}{t} = t, \quad \therefore \lim_{x \rightarrow 0} \int_0^x \arctan t^2 dt = 0$$

Exercises 6.3 The Fundamental Formula of the Definite Integral

1. Find the derivatives of following functions.

(1) $\frac{d}{dx} \int_0^x \sin^2 u \, du = \sin^2 x$

(2) $\frac{d}{dx} \int_0^x \sqrt{1+t^2} dt = \sqrt{1+x^2}$

(3) $\frac{d}{dx} \int_{\sin x}^{\cos x} \sin(\pi t^2) dt = \sin(\pi \cos^2 x) (-\sin x) - \sin(\pi \sin^2 x) \cos x$

$\sin(\pi \cos^2 x) = \sin(\pi - \pi \sin^2 x) = \sin(\pi \sin^2 x)$

2. Find the limits of following functions.

(1) $\lim_{x \rightarrow 0} \frac{1}{x^2} \int_0^x \arctan t^2 dt$

(2) $\lim_{x \rightarrow 0} \frac{(\int_0^x e^t dt)^2}{\int_0^x t^2 e^{t^2} dt}$

$\lim_{x \rightarrow 0} \frac{\arctan(\sin x)^2}{2x} = \lim_{x \rightarrow 0} \frac{(\sin x)^2}{2x \sin x} = 2 \lim_{x \rightarrow 0} \frac{\sin x}{2} = 2$

3. Evaluate the integral.

(1) $\int_0^{\sqrt{e}} \frac{1}{1+x^2} dx$

$= \arctan \sqrt{3} = \frac{\pi}{3}$

(2) $\int_{-2}^1 \frac{1}{\sqrt{1-x^2}} dx = 2 \int_0^{1/2} \frac{1}{\sqrt{1-t^2}} dt$

$= 2 \arcsin t \Big|_0^{1/2} = \frac{\pi}{3}$

(3) $\int_1^9 \sqrt{x} (1+\sqrt{x}) dx$

$= \int_1^9 (x + x^{3/2}) dx = \frac{x^2}{2} + \frac{2}{5} x^{5/2} \Big|_1^9$

$= \frac{65}{2} + \frac{38}{3} = \frac{271}{6}$

(4) $\int_{-1}^0 \frac{3x^4 + 3x^2 + 1}{x^3 + 1} dx$

$= \int_{-1}^0 (3x^2 + \frac{1}{1+x^2}) dx = x^3 \Big|_{-1}^0 + \arctan x \Big|_{-1}^0$

$= 1 + \frac{\pi}{4}$

(5) $\int_0^{\pi} \tan^2 x dx$

$= \int_0^{\pi} (\sec^2 x - 1) dx = \tan x - x \Big|_0^{\pi}$

$= -\pi$

(6) $\int_0^{2\pi} |\sin x| dx$

$= 2 \int_0^{\pi} \sin x dx = -2 \cos x \Big|_0^{\pi} = 4$

$= 4$

(7) $\int_0^2 f(x) dx$, $f(x) = \begin{cases} x+1, & x < 1 \\ x, & x \geq 1 \end{cases}$

$= \int_0^1 (x+1) dx + \int_1^2 x dx = \left(\frac{x^2}{2} + x \right) \Big|_0^1 + \frac{x^2}{2} \Big|_1^2 = \frac{9}{4}$

When $x \geq 0$, $\int_0^x \sin t dt = \int_0^x 1 dt = x$;

When $x < 0$, $\int_0^x \sin t dt = -\int_x^0 \sin t dt = -\int_x^0 (-1) dt = -x$.

$\therefore \int_0^x \sin t dt = |x|$

4. Use the definite integral to estimate the limits of following functions.

(1) $\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n} \right)$;

$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{1}{n} \cdot \frac{1}{1 + \frac{k}{n}} \right)$

$= \int_0^1 \frac{1}{1+x} dx = \ln(1+x) \Big|_0^1 = \ln 2$

(2) $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \dots + \sin \frac{(n-1)\pi}{n} \right)$;

$= \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \left(\sin \frac{k\pi}{n} \cdot \frac{1}{n} \right) = \int_0^1 \sin \pi x dx = -\frac{\cos \pi x}{\pi} \Big|_0^1 = \frac{2}{\pi}$

5. Suppose $f(x)$ is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , $f'(x) \leq 0$. Let

$F(x) = \frac{1}{x-a} \int_a^x f(t) dt$.

Prove that $F'(x) \leq 0$, $x \in (a, b)$.

Proof: $\because f'(x) \leq 0$, $\therefore f(x) \downarrow$, $x \in (a, b)$, $\forall x \in (a, b)$, by mean value theorem,

$F'(x) = \frac{f(x) \cdot (x-a) - \int_a^x f(t) dt}{(x-a)^2} = \frac{f(x)(x-a) - \int_a^x f(t) dt}{(x-a)^2}$, $\exists \xi \in [a, x]$,

$= \frac{f(x) - f(\xi)}{x-a} \leq 0$, i.e., $F'(x) \leq 0$, $x \in (a, b)$.

$$\sin \frac{\pi}{3} = \sqrt{3}/2, \quad \sin \frac{\pi}{4} = \sqrt{2}/2$$

$$-\frac{1}{\sin t} \left(\frac{\pi}{4} \right) = -\left(\frac{2}{\sqrt{3}} - \frac{2}{\sqrt{2}} \right) = -\left(\frac{2}{3}\sqrt{3} - \sqrt{2} \right) = \sqrt{2} - \frac{2}{3}\sqrt{3}$$

Exercises 6.4 Integration by Substitution and Integration by Parts

1. Evaluate the definite integral.

(1) $\int_0^{\pi/2} \sin \phi \cos^3 \phi \, d\phi$:

$$= -\int_0^{\pi/2} \cos^3 \phi \, d(\cos \phi) \\ = -\frac{1}{4} \cos^4 \phi \Big|_0^{\pi/2} = \frac{1}{4}$$

(2) $\int_0^{\sqrt{2}} \sqrt{2-x^2} \, dx$: $x = \sqrt{2} \sin t$

$$= \int_0^{\pi/2} \sqrt{2} \cos t \cdot \sqrt{2} \cos t \, dt \\ = \int_0^{\pi/2} (1 + \cos 2t) \, dt \\ = t \Big|_0^{\pi/2} + \frac{1}{2} \sin 2t \Big|_0^{\pi/2} = \pi/2$$

(3) $\int_1^{\sqrt{e}} \frac{1}{x^2 \sqrt{1+x^2}} \, dx$: $x = \tan t$

$$= \int_{\pi/4}^{\pi/3} \frac{\cot^2 t \cdot \cos t \cdot \sec^2 t \, dt}{\sqrt{1+\tan^2 t}} = \int_{\pi/4}^{\pi/3} \frac{\cot t}{\sec t} \, dt = \int_{\pi/4}^{\pi/3} \cos t \, dt \\ = \sin t \Big|_{\pi/4}^{\pi/3} = \sqrt{2} - \frac{2}{3}\sqrt{3}$$

(4) $\int_1^{1+\sqrt{x}} \frac{1}{1+\sqrt{x}} \, dx$: $\sqrt{x} = t, x = t^2$

$$= \int_1^2 \frac{2t}{1+t} \, dt = \int_1^2 \frac{2(t+1)-2}{1+t} \, dt \\ = 2 \int_1^2 \frac{t}{1+t} \, dt - 2 \int_1^2 \frac{1}{1+t} \, dt \\ = 2 - 2 \ln(t+1) \Big|_1^2 \\ = 2 - 2 \ln \frac{3}{2} = 2 + 2 \ln \frac{2}{3}$$

(5) $\int_0^1 x e^{-x^2} \, dx$:

$$= -\int_0^1 e^{-x^2} d\left(\frac{1}{2}x^2\right) \\ = -e^{-x^2} \Big|_0^1 = 1 - e^{-\frac{1}{2}}$$

(6) $\int_0^1 \frac{1}{x^2 + 2x + 2} \, dx$.

$$= \int_0^1 \frac{d(x+1)}{(x+1)^2 + 1} \\ = \arctan(x+1) \Big|_0^1 \\ = \arctan 2 - \frac{1}{4}\pi$$

$$\tan \frac{\pi}{3} = \tan 60^\circ = \cot 30^\circ = \sqrt{3}$$

$$\therefore \arctan \sqrt{3} = \frac{\pi}{3}$$

2. Evaluate the definite integral.

(1) $\int_1^e \ln x \, dx$:

$$= \int_1^e \ln x \, dx - \int_{1/e}^1 \ln x \, dx \\ = (x \ln x - \int_1^e \frac{1}{x} \, dx) - (x \ln x - \int_{1/e}^1 \frac{1}{x} \, dx) \\ = (e - 1) - [-\frac{1}{e} \ln e - (1 - \frac{1}{e})] \\ = 2 - 2e^{-1}$$

(2) $\int_0^1 \arccos x \, dx$:

$$= x \arccos x \Big|_0^1 - \int_0^1 x \, d(\arccos x) \\ = 0 - \int_0^1 x \frac{-1}{\sqrt{1-x^2}} \, dx \\ = \int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx \\ = -\sqrt{1-x^2} \Big|_0^1 = 1$$

(3) $\int_0^{\sqrt{3}} \arctan x \, d(x^2+1)$

$$= \frac{1}{2} \int_0^3 \arctan x \, d(x^2+1) \\ = \frac{1}{2} (x^2+1) \arctan x \Big|_0^{\sqrt{3}} - \frac{1}{2} \int_0^{\sqrt{3}} \frac{1}{1+x^2} \, dx \\ = \frac{3}{2} \arctan \sqrt{3} - \frac{\pi}{2} \\ = \frac{3}{2} \cdot \frac{\pi}{3} - \frac{\pi}{2} = \frac{\pi}{2}$$

(5) $\int_1^e \sin(\ln x) \, dx = \int_0^1 \sin t \, dt$

$$= x \sin(\ln x) \Big|_1^e - \int_1^e x \cdot \cos(\ln x) \cdot \frac{1}{x} \, dx \\ = e \sin 1 - \int_1^e \cos(\ln x) \, dx \\ = e \sin 1 - x \cos(\ln x) \Big|_1^e + \int_1^e x \sin(\ln x) \cdot \frac{1}{x} \, dx \\ \therefore \int_0^1 \sin t \, dt = \frac{1}{2} (e \sin 1 - e \cos 1 + 1)$$

3. Suppose $f(x)$ is continuous on the closed interval $[a, b]$, prove that

$$\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx.$$

$$\int_a^b f(x) \, dx \stackrel{x=a+b-t}{=} \int_b^a f(a+b-t) \, d(a+b-t) \\ = \int_a^b f(a+b-t) \, dt$$

4. Prove that $\int_1^x \frac{1}{1+t^2} dt = \int_1^x \frac{1}{1+t^2} dt$ ($x > 0$).

$$x > 0, \int_1^x \frac{1}{1+t^2} dt \stackrel{t=1/\theta}{=} \int_{1/x}^1 \frac{1}{1+\theta^{-2}} (-\theta^{-2}) d\theta$$

$$= \int_1^{1/x} \frac{1}{1+\theta^2} d\theta = \int_1^{1/x} \frac{1}{1+t^2} dt$$

5. Prove that $\int_0^1 x^n (1-x)^n dx = \int_0^1 x^n (1-x)^n dx$.

$$\int_0^1 x^n (1-x)^n dx \stackrel{x=1-t}{=} \int_1^0 (1-t)^n t^n d(-t) = \int_0^1 (1-t)^n t^n dt$$

$$= \int_0^1 t^n (1-t)^n dt = \int_0^1 x^n (1-x)^n dx$$

6. Assume $f(x)$ is a continuous function of period T , prove that the value of $\int_a^{a+T} f(x) dx$ is independent of the constant a .

$$\begin{aligned} \int_a^{a+T} f(x) dx &= \int_a^0 f(x) dx + \int_0^T f(x) dx + \int_T^{a+T} f(x) dx \\ &= \int_a^0 f(x) dx + \int_0^T f(x) dx + \int_0^T f(x) dx + \int_0^a f(x) dx \\ &= \int_0^T f(x) dx, \therefore \int_a^{a+T} f(x) dx \text{ is independent of the } a. \end{aligned}$$

7. Assume $f(x)$ is a continuous function. Prove that if $f(x)$ is an odd function, then $\int_0^x f(t) dt$ is an even function; if $f(x)$ is an even function, then $\int_0^x f(t) dt$ is an odd function.

(i) If $f(x)$ is an odd function: $f(-t) = -f(t)$,

$$g(-x) = \int_0^{-x} f(t) dt = \int_0^x f(-t) d(-t) = \int_0^x -f(t) dt = -g(x),$$

$$g(x) = \int_0^x f(t) dt$$

$$g(-x) = \int_0^{-x} f(t) dt = \int_0^x f(-t) d(-t) = -\int_0^x f(t) dt = -g(x),$$

$$g(x) = \int_0^x f(t) dt = -\int_0^x f(t) dt = -g(x),$$

$$g(x) \text{ is an odd function.}$$

(ii) If $f(x)$ is an even function: $f(-t) = f(t)$,

$$g(-x) = \int_0^{-x} f(t) dt = \int_0^x f(-t) d(-t) = \int_0^x f(t) dt = g(x),$$

$$g(x) \text{ is an even function.}$$

书中定理知

$$\frac{x^2(1+x)}{x^2(1+x)} = \frac{a}{x} + \frac{b}{x^2} + \frac{c}{1+x}$$

Exercises 6.6 Improper Integrals

1. Determine the convergence or divergence of the following improper integrals. If it is convergent, please evaluate it.

(1) $\int_1^{+\infty} \frac{2x}{1+x^2} dx$:

divergence.

$$\int_1^{+\infty} \frac{2x}{1+x^2} dx = \int_1^{+\infty} \frac{2x}{1+x^2} dx + \int_0^{+\infty} \frac{2x}{1+x^2} dx$$

$$\int_0^{+\infty} \frac{2x}{1+x^2} dx = \ln(1+x^2) \Big|_0^{+\infty} = +\infty.$$

(2) $\int_e^{+\infty} \frac{1}{x \ln x} dx$:

divergence

$$\int_e^{+\infty} \frac{1}{x \ln x} dx = \ln \ln x \Big|_e^{+\infty} = +\infty$$

(3) $\int_1^{+\infty} \frac{1}{x^2(1+x)} dx$:

convergent.

$$\int_1^{+\infty} \frac{dx}{x^2(1+x)} = \lim_{A \rightarrow +\infty} \int_1^A \left(-\frac{1}{x} + \frac{1}{x^2} + \frac{1}{1+x} \right) dx$$

$$= \lim_{A \rightarrow +\infty} \left(-\frac{1}{x} + \ln \frac{1+x}{x} \right) \Big|_1^A$$

$$= (-\ln 2 + \lim_{A \rightarrow +\infty} (-\frac{1}{A} + \ln \frac{A+1}{A})) = -\ln 2$$

(5) $\int_0^{+\infty} x^n e^{-x} dx$:

convergence

$$\int_0^{+\infty} x^n e^{-x} dx = \Gamma(n+1)$$

$$= n!$$

(6) $\int_1^2 \frac{1}{x \sqrt{x^2-1}} dx$. Convergence

$$\begin{aligned} &= \int_1^2 \frac{x^{-2}}{\sqrt{1-\frac{1}{x^2}}} dx \\ &= -\int_1^2 \frac{d(\frac{1}{x})}{\sqrt{1-(\frac{1}{x})^2}} = -\arcsin \frac{1}{x} \Big|_1^2 \\ &= \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3} \end{aligned}$$

$x \rightarrow 0^+$ or $x \rightarrow 0^-$, $\arctan x \sim x$, $\frac{1}{x^n} \sim x^{-n}$
 $\int_0^1 \frac{1}{x^{n-1}} dx \sim \int_0^1 \frac{1}{x^n} dx \sim \int_0^1 \frac{\arctan x}{x^n} dx \sim \int_0^1 \frac{1}{x^{n+1}} dx$

2. Determine the convergence or divergence of the following integrals.

(1) $\int_0^{+\infty} \frac{x^2}{x^4 - x^2 + 1} dx$: convergence.

(2) $\int_1^{+\infty} \frac{1}{x^3(x^2+1)} dx$: convergence.

$= \int_0^1 \frac{x^2}{x^4 - x^2 + 1} dx + \int_1^{+\infty} \frac{x^2}{x^4 - x^2 + 1} dx$
 $\int_1^{+\infty} \frac{x^2}{x^4 - x^2 + 1} dx = \int_1^{+\infty} \frac{dx}{x^2(x^2+1)}$ and $\int_1^{+\infty} \frac{dx}{x^3(x^2+1)} < +\infty$

$p = \frac{5}{3} > 1$, $\int_1^{+\infty} \frac{dx}{x^{5/3}} < +\infty$

common integral $\int_0^{+\infty} \frac{1}{x^2(x^2+1)} dx$
 "the same of the convergence or divergence"
 about improper integrals $\int_1^{+\infty} \frac{x^2}{x^4 - x^2 + 1} dx$ and $\int_1^{+\infty} \frac{dx}{x^2(x^2+1)} < +\infty$

(3) $\int_0^1 \frac{1}{\ln x} dx$:

(4) $\int_0^{+\infty} \frac{\arctan x}{x^2} dx = \int_0^1 \frac{\arctan x}{x^n} dx + \int_1^{+\infty} \frac{\arctan x}{x^n} dx$

set $f(x) = \frac{1}{\ln x}$, $g(x) = \frac{1}{x-1}$
 $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 1} \frac{x-1}{\ln x} = \lim_{x \rightarrow 1} \frac{1}{1/x} = 1$

When $n-1 < 1$, $\int_0^1 \frac{\arctan x}{x^n} dx$ convergence,
 when $n > 1$, $\int_1^{+\infty} \frac{\arctan x}{x^n} dx$ convergence

$\therefore$ the same of the convergence or divergence about improper integrals
 $\int_1^{+\infty} \frac{1}{x^2} dx$ and $\int_1^{+\infty} \frac{1}{x^{n-1}} dx$ divergence,
 $\therefore \int_1^{+\infty} \frac{1}{x^n} dx$ divergence, for $1 < n < 2$.

(5) $\int_0^1 \frac{\ln x}{1-x^2} dx$.

$f(x) = \frac{\ln x}{1-x^2}$, $x=1$ is removable singular point
 $\lim_{x \rightarrow 1} \frac{\ln x}{1-x^2} = \lim_{x \rightarrow 1} \frac{1/x}{-2x} = -1/2$, $g(x) = \frac{1}{x^{1/2}}$

$\lim_{x \rightarrow 1} \frac{\ln x}{1-x^2} = \lim_{x \rightarrow 1} \frac{1/x}{-2x} = -1/2$, $g(x) = \frac{1}{x^{1/2}}$

$\lim_{x \rightarrow 0^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0^+} \frac{x^{1/2} \ln x}{1-x^2} = \lim_{x \rightarrow 0^+} \frac{\ln x}{x^{-1/2}} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/2 x^{-3/2}} = 0$

$\therefore \int_0^1 g(x) dx$ convergence, $\therefore \int_0^1 \frac{\ln x}{1-x^2} dx$ convergence.

$\int_0^{\pi/4} \ln \cos t dt = \int_{\pi/2}^{\pi/4} \ln \sin(-t) dt = \int_{\pi/4}^{\pi/2} \ln \sin t dt$
 $\therefore \int_0^{\pi/4} \ln \cos t dt = \int_{\pi/4}^{\pi/2} \ln \sin t dt + 2 \int_{\pi/4}^{\pi/2} \ln \sin t dt = \frac{1}{2} \pi \ln 2 + 2 \int_0^{\pi/2} \ln \sin t dt$
 $= \frac{\pi}{2} \ln 2 + 2 \int_0^{\pi/2} \ln \sin t dt$
 $\Rightarrow \int_0^{\pi/4} \ln \cos t dt = \frac{\pi}{4} \ln 2$

3. Evaluate the following improper integrals.

(1) $\int_0^{\pi/2} \ln \sin x dx \hat{=} \int_0^{\pi/2} \ln \cos x dx$

$= \int_0^{\pi/2} \ln(2 \sin \frac{x}{2} \cos \frac{x}{2}) dx$
 $= \int_0^{\pi/2} (\ln 2 + \ln \sin \frac{x}{2} + \ln \cos \frac{x}{2}) dx$
 $= \frac{\pi}{2} \ln 2 + \int_0^{\pi/2} \ln \sin \frac{x}{2} dx + \int_0^{\pi/2} \ln \cos \frac{x}{2} dx$
 $= \frac{\pi}{2} \ln 2 + 2 \int_0^{\pi/4} \ln \sin t dt + 2 \int_0^{\pi/4} \ln \cos t dt$

4. Find the limits of following functions.

(1) $\lim_{x \rightarrow 0} x \left(\int_x^1 \frac{\cos t}{t^2} dt \right)$:

$= \lim_{x \rightarrow 0} \frac{\int_x^1 \frac{\cos t}{t^2} dt}{\frac{1}{x^2}}$
 $= \lim_{x \rightarrow 0} \frac{-\frac{\cos x}{x} - \frac{1}{x^2}}{-\frac{2}{x^3}} = 1$

(2) $\lim_{x \rightarrow +\infty} \frac{\int_0^x \sqrt{1+t^4} dt}{x^3}$

$= \lim_{x \rightarrow +\infty} \frac{\sqrt{1+x^4}}{3x^2} = 1/3$

$\lim_{x \rightarrow 0} \frac{\cos x}{x^2} = \lim_{x \rightarrow 0} \frac{-\sin x}{2x} = \lim_{x \rightarrow 0} \frac{-\cos x}{2} = -1/2$

$$2. \lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{\sin x}{x^3 + 4x^2} = \lim_{x \rightarrow 0} \frac{\cos x}{3x^2 + 8x} = \frac{1}{0} = \infty$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{3x^2 + 4x^3} = 1/3$$

Test 6

I. Choices.

1. Assume $f(x)$ is a continuous function and $F(x) = \int_x^e f(t)dt$, then $F'(x) =$ (A)

(A) $-e^x f(e^x) - f(x)$. (B) $-e^x f(e^x) + f(x)$.

(C) $e^x f(e^x) - f(x)$. (D) $e^x f(e^x) + f(x)$.

2. Let $f(x) = \int_0^{\sin x} \sin t dt$, $g(x) = x^3 + x^4$, if $x \rightarrow 0$, then (B)

(A) $f(x)$ is equivalent to $g(x)$.

(B) $f(x)$ and $g(x)$ are said infinitesimals of the same order but not equivalent.

(C) $f(x)$ is infinitesimal of bigger order than $g(x)$.

(D) $f(x)$ is infinitesimal of smaller order than $g(x)$.

3. Assume

$$M = \int_{-1/2}^{1/2} \frac{\sin x}{1+x^2} \cos^4 x dx = 0, \quad N = \int_{-1/2}^{1/2} (\sin^3 x + \cos^3 x) dx = \int_{-1/2}^{1/2} \cos^3 x dx > 0$$

$P = \int_{-1/2}^{1/2} (x^2 \sin^3 x - \cos^4 x) dx$. Then (D)

(A) $N < P < M$.

(B) $M < P < N$.

(C) $N < M < P$.

(D) $P < M < N$.

4. $F(x) = \int_x^{x^2} e^{\sin t} dt$, then $F'(x) =$ (A)

(A) is a positive constant.

(B) is a negative constant.

(C) $= 0$.

(D) is not a constant.

5. Assume $F(x)$ is a continuous function, which of the following functions is an even function

function

(A) $\int_0^x f(t^2) dt$.

(B) $\int_0^x f^2(t) dt$.

(C) $\int_0^x [f(t) - f(-t)] dt$.

(D) $\int_0^x [f(t) + f(-t)] dt \triangleq F(x)$.

$$F(-x) = \int_0^{-x} [f(t) + f(-t)] dt$$

24

$$u = -t \quad \int_0^x -u [f(-u) + f(u)] (-du) = \int_0^x u [f(u) + f(-u)] du = F(x)$$

$$\int_0^{x+T} f(t) dt = \int_0^x f(t) dt + \int_x^{x+T} f(t) dt$$

II. Calculations and Proof

1. Assume $f(x)$ is a continuous periodic function of period T . Prove that:

$$q(x) \triangleq \int_0^x f(t) dt \text{ is periodic with period } T \Leftrightarrow \int_0^T f(t) dt = 0.$$

Proof. $q(x)$ is periodic with period $T \Leftrightarrow q(x+T) \equiv q(x) \Leftrightarrow$

$$\int_0^{x+T} f(t) dt \equiv \int_0^x f(t) dt \Leftrightarrow \int_x^{x+T} f(t) dt \equiv 0.$$

$$\Leftrightarrow \int_0^{x+T} f(t) dt \equiv \int_0^T f(t) dt \Leftrightarrow \int_0^T f(t) dt = 0$$

(2) Assume $f(x)$ is continuous on the closed interval $[a, b]$ and $f(x) > 0$, prove that:

$$\int_a^b f(x) dx \int_a^b \frac{1}{f(x)} dx \geq (b-a)^2.$$

By Cauchy inequality

$$(b-a)^2 = \left(\int_a^b 1 \cdot dx \right)^2 = \left(\int_a^b \sqrt{f(x)} \cdot \frac{1}{\sqrt{f(x)}} dx \right)^2$$

$$\leq \int_a^b (f(x)) dx \int_a^b \left(\frac{1}{f(x)} \right) dx = \int_a^b f(x) dx \int_a^b \frac{1}{f(x)} dx$$

3. Find $\frac{d}{dx} \int_0^x t(x^2 - t^2) dt$.

$$\therefore \int_0^x t(x^2 - t^2) dt = -\frac{1}{2} \int_0^x (x^2 - t^2) d(x^2 - t^2) = -\frac{1}{2} \int_{x^2}^0 (x^2 - t^2) d(x^2 - t^2)$$

$$= \frac{1}{2} \int_0^{x^2} (x^2 - t^2) dt = \frac{d}{dx} \left(\frac{1}{2} \int_0^{x^2} (x^2 - t^2) dt \right)$$

4. Find $\int_{-1/2}^{1/2} (x^3 + \sin^2 x) \cos^2 x dx$.

$$= 2 \int_0^{1/2} \sin^2 x (1 - \sin^2 x) dx = 2 \int_0^{1/2} \sin^2 x dx - 2 \int_0^{1/2} \sin^4 x dx$$

$$= 2 \left(\int_0^{1/2} \sin^2 x dx - \int_0^{1/2} \sin^4 x dx \right) = 2 \left(\frac{1}{2} \cdot \frac{\pi}{2} - \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \right) = \frac{\pi}{8}$$

$$\int_0^{1/2} \sin^2 x dx = \frac{1}{2} \cdot \frac{\pi}{2}, \quad \int_0^{1/2} \sin^4 x dx = \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

5. Determine the convergence or divergence of the improper integral $\int_1^{\infty} \frac{1}{x(x^2+1)^2} dx$.

$$\because 0 \leq \frac{1}{x(x^2+1)^2} \leq \frac{1}{x^4}, \quad 0 \leq \int_1^{+\infty} \frac{1}{x^4} dx < +\infty$$

$\therefore \int_1^{+\infty} \frac{1}{x(x^2+1)^2} dx$ convergence.

6. Suppose that $f(x)$ is continuous on $[0,1]$ and is differentiable on $(0,1)$.

$f(0) = \int_0^1 e^x f(x) dx$. Prove that: $\exists \xi \in (0,1)$, then $f'(\xi) + f(\xi) = 0$.

Proof. Let $g(x) = e^x f(x)$,

then

$$g(0) = f(0) = \int_0^1 g(x) dx$$

$= g(\xi)(1-0) = g(\xi)$, $\xi \in [0,1]$, by mean value theorem,

$g(x)$ is continuous on $[0,1]$ and is differentiable on

$$(0,1), \quad g(0) = g(\xi),$$

then, there exists $\xi \in (0,1) \subset (0,1)$,

such that

$$g'(\xi) = e^{\xi} f'(\xi) + e^{\xi} f(\xi) = 0,$$

$$\text{i.e. } f'(\xi) + f(\xi) = 0, \quad \xi \in (0,1).$$

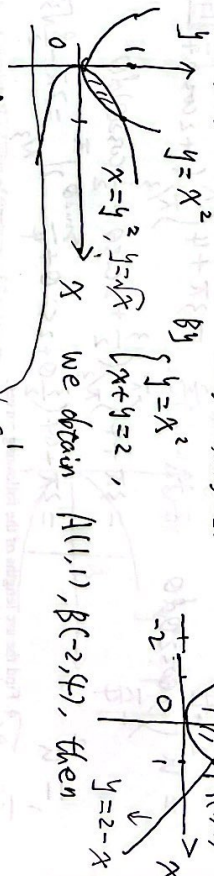
Chapter 7 The Application of Definite Integrals

Exercises 7.2 The Application of Definite Integrals to Geometry

1. Find the area of the plane figure enclosed by the following curves.

(1) $x = y^2, y = x^2$

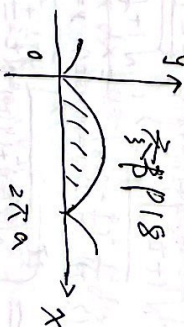
(2) $y = x^2, x + y = 2$.



$$\begin{aligned} S &= \int_0^1 (\sqrt{x} - x^2) dx \\ &= \left(\frac{2}{3} x^{3/2} - \frac{1}{3} x^3 \right) \Big|_0^1 = \frac{1}{3} \end{aligned}$$

$$S = \int_{-2}^1 (2-x-x^2) dx = 9/2$$

2. Find the area enclosed by the x -axis curve and the parametric equation $x = a(t - \sin t), y = a(1 - \cos t) (0 \leq t \leq 2\pi)$.



$$\begin{aligned} P_{2.65} (7.2.6) \\ S &= \int_0^{2\pi} |y(t)x'(t)| dt \end{aligned}$$

$$= \int_0^{2\pi} a^2 (1 - \cos t)^2 dt$$

$$= 4a^2 \int_0^{2\pi} \sin^4 \frac{t}{2} dt$$

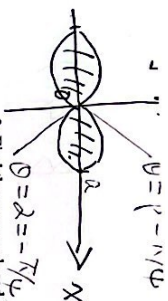
$$= 8a^2 \int_0^{\pi} \sin^4 \frac{t}{2} d\theta$$

$$= 8a^2 \times 2 \int_0^{\pi/2} \sin^4 \theta d\theta$$

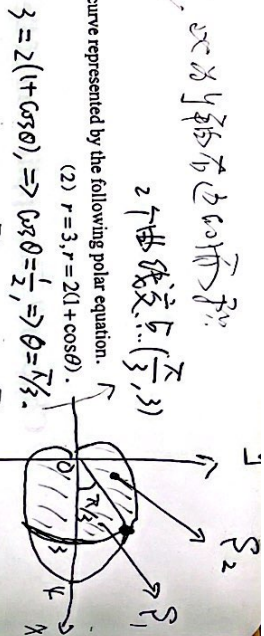
$$\begin{aligned} P_{2.35} \\ &= 8a^2 \times 2 \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = 3\pi a^2 \end{aligned}$$

$$\begin{aligned} \therefore \int_0^{\pi} \sin^4 \theta d\theta &= \int_0^{\pi/2} \sin^4 \theta d\theta + \int_{\pi/2}^{\pi} \sin^4 \theta d\theta \\ &= 2 \int_0^{\pi/2} \sin^4 \theta d\theta \end{aligned}$$

B(-2,4)



3. Find the area enclosed by the curve represented by the following polar equation.



(2.6)(7.2.4) (1) $r^2 = a^2 \cos 2\theta$

$S = 2 \times \frac{1}{2} \int_{-\pi/4}^{\pi/4} r^2(\theta) d\theta$

$S = 2 \times \frac{1}{2} \int_{-\pi/4}^{\pi/4} a^2 \cos 2\theta d\theta$

$= a^2 \int_{-\pi/4}^{\pi/4} \cos 2\theta d\theta$

$= a^2 \left[\frac{1}{2} \sin 2\theta \right]_{-\pi/4}^{\pi/4}$

$= a^2 \left[\frac{1}{2} \sin \frac{\pi}{2} - \frac{1}{2} \sin \left(-\frac{\pi}{2}\right) \right]$

$= a^2 \left[\frac{1}{2} (1) - \frac{1}{2} (-1) \right]$

$= a^2$

4. Find the arc lengths of the following curves.

(1) $y = e^x$ ($0 \leq x \leq \frac{1}{2}$):

$L = \int_0^{1/2} \sqrt{1 + e^{2x}} dx = \int_1^{e^{1/2}} \frac{e^{x/2}}{e^{x/2} + e^{-x/2}} dt$

$= \int_1^{\sqrt{e}} \frac{t}{t^2 + 1} dt = \left[\frac{1}{2} \ln(t^2 + 1) \right]_1^{\sqrt{e}}$

$= \frac{1}{2} \ln(e + 1) - \frac{1}{2} \ln(2)$

(2) $x = a(t - \sin t)$, $y = a(1 - \cos t)$ ($0 \leq t \leq 2\pi$):

$L = \int_0^{2\pi} \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt = a \int_0^{2\pi} \sqrt{2 - 2\cos t} dt$

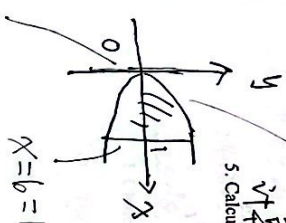
$= a \int_0^{2\pi} 2 \sin \frac{t}{2} dt = 4a \left[-\cos \frac{t}{2} \right]_0^{2\pi} = 4a(1 - 1) = 0$

$L = \int_0^{2\pi} \sqrt{r^2(\theta) + (r'(\theta))^2} d\theta = a \int_0^{2\pi} \sqrt{1 + \theta^2} d\theta$

$= a \left[\frac{1}{2} \theta \sqrt{1 + \theta^2} + \frac{1}{2} \ln(\theta + \sqrt{1 + \theta^2}) \right]_0^{2\pi}$

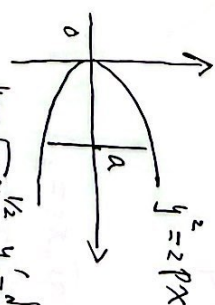
$= a \left[\pi \sqrt{1 + 4\pi^2} + \frac{1}{2} \ln(2\pi + \sqrt{1 + 4\pi^2}) \right]$

$y = 2\sqrt{x} = f(x)$



$V = \pi \int_a^b f^2(x) dx = \pi \int_0^1 4x dx = 2\pi$

5. Calculate the volume of the revolution of a parabola $y^2 = 4x$ and a straight line $x = 1$.



$y = \sqrt{2p}x^{1/2}$, $y' = \frac{\sqrt{2p}}{2}x^{-1/2}$

6. Calculate the Areas of Surfaces of Revolution.

(1) $y^2 = 2px$ ($0 \leq x \leq a$), about the x-axis and the y-axis:

(i) $A_x = 2\pi \int_0^a y(x) \sqrt{1 + (y'(x))^2} dx = 2\pi \int_0^a \sqrt{x} \sqrt{1 + \frac{p}{x}} dx$

$= \frac{4}{3} \pi \sqrt{2p} \left[x + \frac{1}{2} p \right]_0^a = \frac{4}{3} \pi \sqrt{2p} \left[a + \frac{1}{2} p \right]$

(ii) $A_y = 2 \times 2\pi \int_0^a x \sqrt{1 + (y'(x))^2} dx = 4\pi \int_0^a x \sqrt{x + \frac{p}{2}} dx$

$= 4\pi \left[\frac{1}{2} (x + \frac{p}{2}) \sqrt{x + \frac{p}{2}} - \frac{1}{2} \times \frac{1}{2} p \sqrt{x + \frac{p}{2}} \right]_0^a$

$= \frac{\pi}{4} \left[(4a + p) \sqrt{2a + p} - p \sqrt{p} \right]$

$A_x = 2\pi \int_0^{2\pi} a(1 - \cos t) \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt$

$= 2\pi \int_0^{2\pi} a(1 - \cos t) \sqrt{2 - 2\cos t} dt = 8\pi a \int_0^{2\pi} \sin \frac{t}{2} dt$

$= \frac{64}{3} \pi a^2$

$A_y = 2\pi \int_0^{2\pi} a(t - \sin t) \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt$

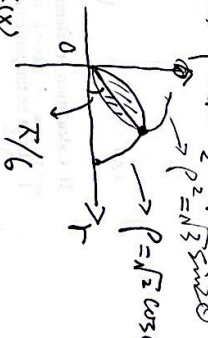
$= 4\pi a^2 \int_0^{2\pi} (t - \sin t) \sin \frac{t}{2} dt = 16\pi a^2$

Exercises 7.3 The Application of Definite Integrals in Physics

- Find the length and center of mass of the star line $x = a \cos^3 t$, $y = a \sin^3 t$ ($0 \leq t \leq \pi$).

- Suppose a uniform disk of mass M , and radius R , find the moment of inertia of the disk with respect to one of its diameters.

30

3. 求弧长, $\rho' = \sqrt{3} \sin 2\theta = \sqrt{3} \cdot \frac{1}{2} \cdot 2 \sin \theta \cos \theta = \sqrt{3} \sin \theta$
 $\Rightarrow \rho = \sqrt{6} \sin \theta = \sqrt{2} \sin 2\theta \Rightarrow \tan \theta = \frac{\sqrt{3}}{3}, \Rightarrow \theta = \pi/6, \rho = \sqrt{2} \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$
 $\therefore$ 两曲线交点为 $(\frac{\sqrt{3}}{2}, \sqrt{3}/2)$
 由 $\rho = 2\cos\theta$ 和 $\rho = \sqrt{2} \sin 2\theta$ 求交点, 本起因中 $\rho = 2\cos\theta$
 $= \int_0^{\pi/6} \frac{1}{2} \sqrt{3} \sin 2\theta d\theta + \int_{\pi/6}^{\pi/2} \frac{1}{2} \sqrt{3} \sin 2\theta d\theta$
 Test 7
 1. Choices $= \frac{1}{2} \int_{\pi/6}^{\pi/2} 2 \cos^3 \theta d\theta$


- The area of graph enclosed by the curve $y = x(x-1)(x-2)$ and x -axis can be expressed as

(A) $-\int_0^2 x(x-1)(2-x)dx$

(B) $\int_0^1 x(x-1)(2-x)dx - \int_1^2 x(x-1)(2-x)dx$

(C) $-\int_0^1 x(x-1)(2-x)dx + \int_1^2 x(x-1)(2-x)dx$

(D) $\int_0^2 x(x-1)(2-x)dx$

- The area of the surface generated by revolving the curve $y^2 = 2px$ ($0 \leq x \leq a$) about the x -axis can be expressed as

(A) $2\pi \int_0^a \sqrt{2px} \sqrt{p+2x} dx$

(B) $2\pi \int_0^a \sqrt{2px} dx$

(C) $2\pi \int_0^a \sqrt{2px} \sqrt{p+2x} dx$

(D) $2\pi \int_0^a \sqrt{\frac{p+2x}{2x}} dx$

- The area of the common part enclosed by the curves $\rho = \sqrt{2} \cos \theta$ and $\rho^2 = \sqrt{3} \sin 2\theta$ is

(A) $\frac{1}{2} \int_0^{\pi/6} 2 \cos^2 \theta d\theta + \frac{1}{2} \int_{\pi/6}^{\pi/2} \sqrt{3} \sin 2\theta d\theta$

(B) $\frac{1}{2} \int_0^{\pi/6} \sqrt{3} \sin 2\theta d\theta + \frac{1}{2} \int_{\pi/6}^{\pi/2} 2 \cos^2 \theta d\theta$

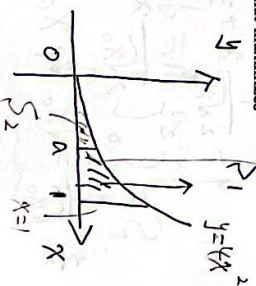
(C) $\frac{1}{2} \int_{\pi/6}^{\pi/2} \sqrt{3} \sin 2\theta d\theta - \frac{1}{2} \int_0^{\pi/6} 2 \cos^2 \theta d\theta$

(D) $\frac{1}{2} \int_0^{\pi/6} \sqrt{3} \sin 2\theta d\theta + \frac{1}{2} \int_{\pi/6}^{\pi/2} 2 \cos^2 \theta d\theta$

- Let S_1 be the part bounded by $y = 4x^2$ and $x = a$, $x = 1$, $y = 0$, S_2 be the part bounded by $y = 4x^2$ and $x = a$, $x = 1$, $y = 0$ ($0 < a < 1$). V_1, V_2 are the volumes of the rotation of S_1, S_2 respectively, what is the value of a that maximizes

$V_1 = \pi \int_0^a (4x^2)^2 dx = \pi \int_0^a 16x^4 dx = \frac{16\pi}{5} a^5$
 $V_2 = \pi \int_a^1 (4x^2)^2 dx = \frac{16\pi}{5} (1 - a^5)$
 $V(a) = V_1 + V_2 = \pi (2a + \frac{16}{5} - \frac{16}{5} a^5)$
 $V'(a) = \pi (2 - 16a^4) = 0 \Rightarrow a = \frac{1}{2}$
 $V(\frac{1}{2}) = \frac{16\pi}{5}$

S_1 绕 x 轴旋转...
 S_2 绕 y 轴旋转...
 $V_1 = \pi \int_0^a (4x^2)^2 dx$
 $V_2 = \pi \int_a^1 (4x^2)^2 dx$
 $V(a) = V_1 + V_2 = \pi (2a + \frac{16}{5} - \frac{16}{5} a^5)$
 $a = 1/2, V(1/2) = \frac{16\pi}{5}$



(A) 1.

$$\sqrt{B}/2.$$

(C) $\frac{1}{3}$.

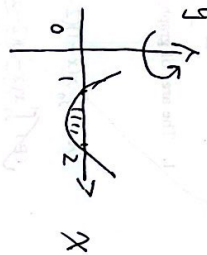
(D) $\frac{1}{4}$.

II. Calculation problems

1. Let A be the part bounded by the curve $y = (x-1)(x-2)$ and x -axis, calculate the volume of the revolution of A about y -axis.

P273 (7.2.15)

$$V = 2\pi \int_1^2 x|(x-1)(x-2)|dx$$



$$= -2\pi \int_1^2 x(x-1)(x-2)dx$$

$$= -2\pi \int_0^1 (t+1)t(t-1)dt = -2\pi \int_0^1 (t^3-t)dt = \pi/2$$

2. Find the length of the curve $y = \ln(1-x^2)$ ($0 \leq x \leq \frac{1}{2}$).

P268 (7.2.9)

$$L = \int_0^{1/2} \sqrt{1+(y'(x))^2} dx = \int_0^{1/2} \sqrt{1+(-\frac{2x}{1-x^2})^2} dx$$

$$= \int_0^{1/2} \frac{1+x^2}{1-x^2} dx = \int_0^{1/2} \frac{2}{1-x^2} dx - \int_0^{1/2} \frac{1-x^2}{1-x^2} dx = \ln \frac{1+x}{1-x} \Big|_0^{1/2} - \frac{1}{2} = \ln 3 - \frac{1}{2}$$

$$\frac{1}{\sqrt{1-x^2}} \geq \frac{1}{\sqrt{1-x^2}} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\frac{1}{\sqrt{1-x^2}} \geq \frac{1}{\sqrt{1-x^2}} \cdot \frac{1}{\sqrt{1-x^2}} = 1.$$

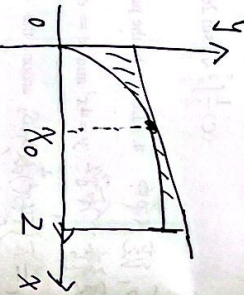
3. Find a tangent line l of the curve $y = \sqrt{x}$, which makes the tangent line, the curve and

straight line $x = 0, x = 2$ formed by the plane figure of the smallest.

$$y = \sqrt{x}, y' = \frac{1}{2\sqrt{x}}$$

$$y - \sqrt{x_0} = \frac{1}{2\sqrt{x_0}}(x - x_0)$$

$$y = \frac{1}{2\sqrt{x_0}}x + \frac{1}{2}\sqrt{x_0}$$



$$S = \int_0^2 \left(\frac{1}{2\sqrt{x_0}}x + \frac{1}{2}\sqrt{x_0} - \sqrt{x} \right) dx = \frac{1}{4\sqrt{x_0}}x^2 + \frac{1}{2}\sqrt{x_0}x - \frac{2}{3}x^{3/2} \Big|_0^2$$

$$= \frac{1}{2\sqrt{x_0}} + \sqrt{x_0} - \frac{2}{3}\sqrt{2}.$$

4. Find a point M on a curve $r^2 = 4\cos 2\theta$ ($0 \leq \theta \leq \frac{\pi}{4}$), that OM bisects the area of the

plane bounded by the curve and the polar axis.

P263 (7.2.4)

$$S_1 = \frac{1}{2} \int_0^{\pi/4} r^2 d\theta = 2 \int_0^{\pi/4} \cos 2\theta d\theta$$

$$= \sin 2\theta \Big|_0^{\pi/4} = 1,$$

(7.2.4)

$$S_2 = \frac{1}{2} \int_{\pi/4}^{\pi/2} r^2 d\theta = \frac{1}{2} \int_{\pi/4}^{\pi/2} \cos 2\theta d\theta = \frac{1}{2} \int_{\pi/4}^{\pi/2} \cos 2\theta d\theta = \frac{1}{2} \sin 2\theta \Big|_{\pi/4}^{\pi/2} = \frac{1}{2}$$

$$\theta_0 = \frac{\pi}{12}, r = \sqrt{4\cos 2\theta} = 2\sqrt{\cos 2\theta} = 2\sqrt{\cos 30^\circ} = 2\sqrt{\frac{\sqrt{3}}{2}} = \sqrt{2\sqrt{3}}$$

$$\theta = \frac{\pi}{12}, r = \sqrt{4\cos 2\theta} = 2\sqrt{\cos 30^\circ} = 2\sqrt{\frac{\sqrt{3}}{2}} = \sqrt{2\sqrt{3}}$$

5. Let A be the area of the trapezoid enclosed by the lines $y = ax + b, x = 1, x$ -axis and y

-axis, find a, b to minimize the volume of the trapezoid about x -axis, in which

$$a \geq 0, b \geq 0.$$

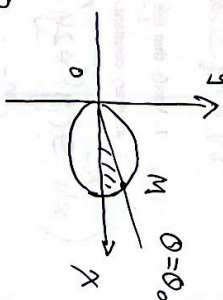
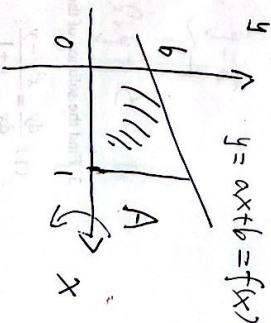
$$A = \int_0^1 (ax + b) dx = \frac{1}{2}a + b, \Rightarrow b = A - \frac{1}{2}a$$

$$V = \pi \int_0^1 (ax + b)^2 dx$$

$$= \pi \int_0^1 (a^2x^2 + 2abx + b^2) dx$$

$$= \pi \left(\frac{1}{3}a^2 + ab + b^2 \right) = \pi \left(\frac{1}{3}a^2 + A^2 \right),$$

$$\frac{1}{3}a = 0 \text{ when } V_{\min} = \pi A^2, \text{ when } b = A, a = 0.$$



Chapter 8 Differential equation

Exercises 8.1 & 8.2 The Concept of Differential equations & Several

kinds of simple Differential Equations

1. Verify that the function $y = Ce^x$ is the solution of the equation $y'' - 2y' + y = 0$ (C is an arbitrary constant).

$$\therefore y'' - 2y' + y = Ce^x - 2Ce^x + Ce^x = 0,$$

$\therefore y = Ce^x$ is the solution of the equation

$$y'' - 2y' + y = 0$$

2. Solve the initial value problem $y' + y = 0$, $y(3) = 2$. If the general solution is $y = Ce^{-x}$, in which C is an arbitrary constant.

$$y = Ce^{-x}, y(3) = 2, \Rightarrow C = 2e^3,$$

$$\therefore y = 2e^{3-x}.$$

3. Find the solution of the following differential equations.

$$(1) \frac{dy}{dx} = \frac{x^2 y - y}{y+1};$$

$$(2) x \frac{dy}{dx} = y + \sqrt{x^2 - y^2};$$

$$y \neq 0 \text{ or } y = 0$$

$$\frac{dy}{dx} = \frac{y}{y+1}(x^2-1), \int \frac{y+1}{y} dy = \int (x^2-1) dx$$

$$\therefore u + x \frac{du}{dx} = u + \sqrt{1-u^2} \sin x, \quad \frac{du}{dx} = \frac{u}{x} + \sqrt{1-\left(\frac{y}{x}\right)^2} \sin x, \quad \text{let } u = \frac{y}{x}$$

$$x \frac{du}{dx} = \sqrt{1-u^2} \sin x, \quad \int \frac{du}{\sqrt{1-u^2}} = \int \frac{\sin x}{x} dx,$$

$$\arcsin \frac{y}{x} = \sin x \cdot \ln|x| + C, \quad \frac{y}{x} = \sin x$$

$$(3) \frac{dy}{dx} = \frac{x-y+5}{x-y-2};$$

$$(4) y'' - y' - x = 0;$$

$$x-y=u, \text{ let } \frac{du}{dx} = 1 - \frac{dy}{dx}$$

$$y' - y = \frac{1}{2}x^2 + C_1$$

$$\Rightarrow \frac{1}{u-2} = -\frac{1}{u-2}, \Rightarrow y = e^x \left(C_2 + \int \left(\frac{1}{2}x^2 + C_1 \right) e^{-x} dx \right)$$

$$(u-2)du = -7dx, \quad = C_2 e^x + e^x \left(-\int \left(\frac{1}{2}x^2 + C_1 \right) de^{-x} \right)$$

$$\int (u-2)dx = -7 \int dx, \quad = C_2 e^x + e^x \left[-e^{-x} \left(\frac{1}{2}x^2 + C_1 \right) + \int x e^{-x} dx \right]$$

$$\therefore \frac{1}{2}u^2 - 2u = -7x + C, \quad = C_2 e^x - \frac{1}{2}x^2 - C_1 - x - 1, \quad y = C_2 e^x - \frac{1}{2}x^2 - x + C$$

$$(5) xy'' + (y')^2 = 0.$$

$$\text{let } y' = p = \frac{dy}{dx}, \text{ let } y'' = \frac{dp}{dx} = \frac{dp}{dy} \frac{dy}{dx} = p \frac{dp}{dy}$$

$$\therefore xy \frac{dp}{dy} + p^2 = 0, \quad \text{let } p = 0, y = C.$$

$$\text{let } y \frac{dp}{dy} + p = 0 \Rightarrow \frac{dp}{p} = -\frac{dy}{y} \Rightarrow \ln p = -\ln y + C$$

$$\Rightarrow p = e^{-\ln y} C = \frac{1}{y} \times \frac{1}{2} C_1$$

$$\Rightarrow 2y dy = C_1 dx, \quad \int 2y dy = \int C_1 dx + C_2$$

$$y^2 = C_1 x + C_2$$

$$(y = C_1 x + C_2) \Rightarrow y^2 = C_1 x + C_2$$

注意: (8.3.4) 中指出, $\int \sin x \cos x e^{\sin x} dx = \int \sin x d e^{\sin x}$
 该式中的 $\sin x$ 不定积分? $\int \sin x e^{\sin x} dx = \int \sin x d e^{\sin x} = \sin x e^{\sin x} - \int e^{\sin x} d \sin x = (\sin x - 1) e^{\sin x}$
 为原函数的一个不定积分, $= \sin x e^{\sin x} - e^{\sin x} = (\sin x - 1) e^{\sin x}$
 不是第 102 页 C.

Exercises 8.3 First-order differential equations

1. Solve the following equations.

(1) $y' + y \cos x = e^{2x}$:

(8.3.4) $y' + y \cos x = e^{2x}$
 $y = e^{-\int \cos x dx} \left(C + \int e^{2x} \cos x dx \right)$
 $y = e^{-\sin x} \left(C + \int \frac{1}{2} \sin 2x e^{\sin x} dx \right)$

$= e^{-\sin x} \left(C + \int \frac{\sin x}{2} (e^{\sin x} + e^{-\sin x}) dx \right)$
 $= e^{-\sin x} \left(C + \frac{\sin x}{2} e^{\sin x} - \frac{\sin x}{2} e^{-\sin x} \right)$

$\int \sin x \cos x e^{\sin x} dx = \int \sin x (1 - \sin^2 x) e^{\sin x} dx$
 $= -\frac{1}{3} \sin^3 x e^{\sin x} + \frac{1}{2} \sin x e^{\sin x} - \frac{1}{2} \sin x e^{-\sin x}$

$y' + \tan x \cdot y = \frac{1}{\cos x}$, $y = e^{-\int \tan x dx} \left(C + \int \frac{1}{\cos x} e^{\tan x} dx \right)$

$= e^{-\ln \cos x} \left(C + \int \frac{1}{\cos x} e^{-\ln \cos x} dx \right)$
 $= \cos x \left(C + \int \frac{1}{\cos^2 x} dx \right)$
 $= \cos x (C + \tan x) = \sin x + C \cos x$

2. Find solutions to the following initial value problems.

(1) $y' + \frac{y}{x} = \frac{\sin x}{x}$, $y(\pi) = 1$:

$y = e^{-\int \frac{1}{x} dx} \left(C + \int \frac{\sin x}{x} e^{\frac{1}{x}} dx \right)$
 $y = e^{-\frac{1}{x}} \left(C - \int e^x \sin x dx \right)$

$= \frac{1}{x} (C - \int x e^x dx)$
 $= \frac{1}{x} (C - (x-1)e^x)$
 $\Rightarrow y(1) = 0, \Rightarrow C = 0, \therefore y = \frac{1-x}{x} e^x$

$\therefore y = \frac{\pi-1-\cos x}{x}$
 $= x e^x - \int e^x dx = x e^x - e^x = (x-1) e^x$

3. Find the solution of the following differential equations.

(1) $3y^2 y' - ay^3 = x+1$:

$y^3 = u, \therefore \frac{du}{dx} = 3y^2 \frac{dy}{dx} = 3y^2 y'$

$\therefore \frac{du}{dx} - au = x+1$

$\therefore P_{297}(8.3.4) u = e^{\int a dx} \left(C + \int (x+1) e^{-ax} dx \right)$

$= e^{ax} \left(C + \int (x+1) e^{-ax} dx \right) = e^{ax} \left(C - \frac{1}{a} \int (x+1) d e^{-ax} \right)$
 $= e^{ax} \left(C - \frac{1}{a} (x+1) e^{-ax} - \frac{1}{a^2} e^{-ax} \right)$

$\therefore \frac{1}{3} \frac{du}{dx} = \frac{1}{3} \frac{d}{dx} (y^3) = y^2 y' = C e^{ax} - \frac{1}{a} x - \frac{a+1}{a^2}$
 $(2) y' \sin x = (\sin x - y^2) \cos x$

$y^2 = u, \therefore \frac{du}{dx} = 2y \frac{dy}{dx} = 2y y'$

$\sin x \frac{du}{dx} = 2(\sin x - u) \cos x$

$\frac{du}{dx} + 2 \cot x \cdot u = 2 \cos x$

$u = e^{-\int 2 \cot x dx} \left(C + \int 2 \cos x e^{\int 2 \cot x dx} dx \right)$

$= e^{-2 \ln \sin x} \left(C + 2 \int \cos x \cdot \sin^2 x dx \right)$

$= \frac{1}{\sin^2 x} \left(C + \frac{2}{3} \sin^3 x \right)$
 $\Rightarrow \frac{1}{\sin^2 x} y^2 = \frac{C}{\sin^2 x} + \frac{2}{3} \sin x$

$\int e^{\cot x} dx = \int \frac{\cos x}{\sin x} dx = \int \frac{d \sin x}{\sin x} = \ln \sin x$

Exercises 8.4 & 8.5 Exact differential equation and the integrating factor method & Second-order linear differential equations with constant coefficients

Find the general solution of the following equations.

(1) $\frac{dy}{dx} - \frac{n}{x}y = e^x x^n$, n is a constant;

(2) $\frac{dy}{dx} = \frac{y}{x+y^3}$, $y \neq 0$ or $x \neq 0$

(8.3.4) $y' = e^{\frac{n}{x}dx} (c + \int x^n e^{-x} \frac{dx}{dx})$
 $= x^n (c + \int e^x dx) = x^n (c + e^x)$

$\therefore x = e^{\int \frac{1}{x} dx} (c + \int y^3 \cdot e^{\frac{1}{x}} dx)$
 $= y(c + \int y dy) = y(c + \frac{1}{2}y^2)$

$\therefore \frac{1}{2}y^3 = c + \frac{1}{2}y^2$
 $\frac{y}{2} \text{ 求解 } y = 0.$

$\frac{y}{2} \text{ 求解 } y = 0.$

(3) $x \frac{dy}{dx} + y = x^3$

(4) $(x + \sin y)dx + (x \cos y - 2y)dy = 0$

$\frac{dy}{dx} + \frac{1}{x}y = x^2$

$y = e^{-\int \frac{1}{x} dx} (c + \int x^2 e^{\frac{1}{x}} dx)$

$= \frac{1}{x} (c + \int x^3 dx)$

$= \frac{c}{x} + \frac{1}{4}x^3$

$\therefore \text{ 通解为 } y = \frac{c}{x} + \frac{1}{4}x^3$

38

P297

(8.3.4) (5) $y' = 2xy - x$

$y' - 2xy = -x$

$y = e^{\int 2x dx} (c - \int x e^{-2x} dx)$

$= e^{x^2} (c - \int x e^{-x^2} dx)$

$= e^{x^2} (c + \frac{1}{2} e^{-x^2})$

$= \frac{1}{2} + c e^{x^2}$

(6) $y'' - y' - 30y = 0$

$\lambda^2 - \lambda - 30 = 0$

$(\lambda + 5)(\lambda - 6) = 0$

$\lambda_1 = -5, \lambda_2 = 6$

$\therefore \text{ 通解为 } (8.4.4) \text{ 通解}$

$y = c_1 e^{-5x} + c_2 e^{6x}$

(7) $y'' - y' - 2y = 4x^2$

$y'' - y' - 2y = 0$

$\lambda^2 - \lambda - 2 = 0, (\lambda + 1)(\lambda - 2) = 0$

$\lambda_1 = -1, \lambda_2 = 2, \text{ 由 P297 (8.3.4)}$

$\text{ 通解为 } y = c_1 e^{-x} + c_2 e^{2x}$

$\text{ 由 P303 表格 (0 不是特征根)}$

$\text{ 设 } y'' - y' - 2y = 4x^2 \text{ 的一个特解为}$

$y_0 = ax^2 + bx + c, \text{ 代入方程}$

$\text{ 求得 } a = -2, b = 2, c = -3$

$\therefore y_0 = -2x^2 + 2x - 3$

 通解为

$y = y_0 + y_1 = c_1 e^{-x} + c_2 e^{2x} - 2x^2 + 2x - 3$

(8) $y'' - y' - 2y = 8 \sin 2x$

$y'' - y' - 2y = 0, \lambda^2 - \lambda - 2 = 0,$

$\lambda_1 = -1, \lambda_2 = 2, \text{ 由 P301 (8.4.4) 通解}$

$\text{ 通解为 } y = c_1 e^{-x} + c_2 e^{2x}$

 由 P303 表格 通解

$y'' - y' - 2y = 8 \sin 2x \text{ 的一个特解为}$

$y_0 = a \cos 2x + b \sin 2x$

 代入方程求得

$a = \frac{2}{5}, b = -\frac{6}{5}$

$\therefore y_0 = \frac{2}{5} \cos 2x - \frac{6}{5} \sin 2x$

 通解为

$y = y_0 + y_1 = c_1 e^{-x} + c_2 e^{2x} + \frac{2}{5} \cos 2x - \frac{6}{5} \sin 2x$

2. $y = cx + f(c)$, $y' = c$, $\lambda = \mp 1$, $y = cx + f(c) = xy' + f(y')$
 $\therefore y = cx + f(c)$ 是 $y = xy' + f(y')$ 的解. $y = cx + f(c)$ 又是 $y = cx + f(c)$ 的解.
 通解: 设 $y = cx + f(c) + C_1$ Test 8 是 $y = cx + f(c)$ 的解, $y = c$,
 1. Choices $\Rightarrow C_1 + cx + f(c) = y = xy' + f(y') = cx + f(c) \Rightarrow C_1 = 0$
 1. The equation $x(y')^2 + y^2 - x = 0$ is (B)

- (A) Second order differential equations. (B) First order differential equations. $\therefore y = cx + f(c)$
 (C) First order linear differential equations. (D) Second order linear differential equations. $\therefore y = cx + f(c)$
 2. If the differential equation is $y = xy' + f(y')$, then the function $y = cx + f(c)$ is (A, B) 是原方程的通解.
 (A) the solution to this equation. (B) the general solution to this equation.
 (C) a particular solution of this equation. (D) not the solution to this equation.

3. The second order homogeneous differential equation with constant coefficients with a particular solution $y_1 = 2e^x, y_2 = 3e^x$ is (A)
 (A) $y'' - y = 0$. (B) $y'' - y' - y = 0$.
 (C) $y'' + y = 0$. (D) $y'' + y' = 0$.
 4. Let $y_1(x), y_2(x), y_3(x)$ be solutions to second order nonhomogeneous linear differential equations $y'' + p(x)y' + q(x)y = f(x)$ which are linearly independent functions, C_1, C_2 are arbitrary constants, then the general solution of the inhomogeneous equation is

- (D) $C_1 y_1 + C_2 y_2 + (1 - C_1 - C_2) y_3$
 (A) $C_1 y_1 + C_2 y_2 + y_3 = C_1(y_1 - y_3) + C_2(y_2 - y_3) + y_3$
 (B) $C_1 y_1 + C_2 y_2 - (C_1 + C_2) y_3$
 (C) $C_1 y_1 + C_2 y_2 - (1 - C_1 - C_2) y_3$
 (D) $C_1 y_1 + C_2 y_2 + (1 - C_1 - C_2) y_3$
 $\therefore y'' + p(x)y' + q(x)y = f(x)$ 的通解.
 5. A particular solution of differential equation $y'' - y = e^x + 1$ should have the form $y = f(x) + g(x)$
 考虑 $y'' - y = e^x$, $a = 1, y'' - y = 0$ 的通解为 (B)
 (A) $ae^x + b$. (B) $ae^x + b$. (C) $ae^x + bx$. (D) $axe^x + bx$.

将 $\lambda^2 - 1 = 0, \lambda = \pm 1, a = 1$ 代入特征方程, 由特征根求特解, 齐次方程
 (将 $\lambda \in \mathbb{R}, \lambda = a \neq 0, a$ 为常数) $\Rightarrow y = e^{ax}$ 是 $y'' - y = 0$ 的通解, $Q_0(x)$ 是 $y'' - y = 0$ 的通解, $\therefore y = e^{ax}$ 是 $y'' - y = 0$ 的通解.
 $\therefore y = e^{ax}$ 是 $y'' - y = 0$ 的通解. 又 $y'' - y = 1$ 的通解为 $y = -1$ (齐次)
 于是 $y'' - y = e^x + 1$ 的通解为 $y = e^x - 1$.

II. Calculation problems

1. Find the general solution of the differential equation $y'' - y' + 4y = 0$.

特征方程 $\lambda^3 - \lambda^2 + 4\lambda - 4 = 0$, 解得 $\lambda_1 = 1, \lambda_2 = 2i$,
 $\lambda_3 = -2i$, 由 P 301 ~ 302 页
 齐次方程 $y = C_1 e^x + C_2 \cos 2x + C_3 \sin 2x$

2. If $f(x)$ satisfies the condition $\begin{cases} y'' + 4y' + 4y = 0 \\ y(0) = 2, y'(0) = -4 \end{cases}$, find the generalized integral

$$\int_0^{+\infty} f(x) dx.$$

$\lambda^2 + 4\lambda + 4 = 0, \lambda_{1,2} = -2$
 由 P 302 页, 齐次 $y'' + 4y' + 4y = 0$ 的通解为
 $y = (C_1 + C_2 x) e^{-2x}, y' = [C_2 - 2(C_1 + C_2 x)] e^{-2x}$
 由 $y(0) = 2, y'(0) = -4$ 得 $C_1 = 2, C_2 = 0, \therefore f(x) = 2e^{-2x}$
 $\int_0^{+\infty} f(x) dx = \int_0^{+\infty} 2e^{-2x} dx = -e^{-2x} \Big|_0^{+\infty} = 1$

3. Set $f(x) = e^x - \int_0^x (x-t)f(t) dt$, in which $f(t)$ is a continuous function, find $f(x)$.

$f(x) = e^x - \int_0^x f(t) dt + \int_0^x f(t) dt$, 由 $f(0) = 1, f'(x) = e^x - \int_0^x f(t) dt - x f(x)$
 $+ x f(x) = e^x - \int_0^x f(t) dt$. $f'(0) = 1, f''(x) = e^x - f(x)$, 即 $f''(x) + f(x) = e^x$
 设 $y = f(x)$, 则 $y'' + y = e^x$. 特征方程 $\lambda^2 + 1 = 0, \lambda = \pm i$, 由 P 302 页
 $y'' + y = 0$ 的通解为 $C_1 \cos x + C_2 \sin x$. 由 P 303 页求特解 $y'' + y = e^x$ 的通解为
 $C e^x$ (又易知 $\frac{1}{2} e^x$ 是 $y'' + y = e^x$ 的一个特解). $\therefore y'' + y = e^x$ 的通解为
 $y = f(x) = C_1 \cos x + C_2 \sin x + \frac{1}{2} e^x$. 由 $f(0) = f'(0) = 1$ 可解出 $C_1 = C_2 = \frac{1}{2}$
 $\therefore f(x) = \frac{1}{2} (\cos x + \sin x) + \frac{1}{2} e^x$.

4. If $f(x), g(x)$ satisfy $f'(x) = g(x), g'(x) = 2e^x - f(x)$, and $f(0) = 0, g(0) = 2$, find

$$\int_0^x \frac{g(x) - f(x)}{(1+x)^2} dx.$$

$$f'(x) = g(x) \Rightarrow f'(x) = 2e^x - f(x), \therefore f''(x) + f(x) = 2e^x$$

$$\text{特征方程 } \lambda^2 + 1 = 0, \lambda = \pm i, \text{ 齐次解 } f_h(x) = C_1 \cos x + C_2 \sin x$$

$$f(0) = 0, f'(0) = 2 \Rightarrow C_1 = 0, C_2 = 2$$

$$\int_0^x \left(\frac{g(x) - f(x)}{(1+x)^2} \right) dx = \int_0^x \frac{f'(x) - f(x)}{(1+x)^2} dx = \int_0^x \left(\frac{f(x)}{(1+x)} \right)' dx$$

5. Suppose the function $f(x)$ has a second continuous derivative, and $f'(0) = 0$.

$$f(x) = e^x - \frac{1}{2} \int_0^x (f''(t) + f'(t) + t) dt.$$

$$f(0) = 1, f'(0) = 0, \therefore \int_0^x [f''(t) + f'(t) + t] dt = \int_0^x (f'(t) + f(t))' dt + \int_0^x t dt$$

$$f(x) = e^{-x} - \frac{1}{2} [f(x) - f(0)] - \frac{1}{2} [f(x) - f(0)] - \frac{1}{4} x^2$$

$$f(x) = e^{-x} - \frac{1}{2} f(x) - \frac{1}{4} x^2 + \frac{1}{2}, \text{ 即 } f'(x) + f(x) = 2e^{-x} - \frac{1}{2} x^2 + 1$$

$$f(x) = e^{-3x} \left(C + \int (2e^{-x} - \frac{1}{2} x^2 + 1) e^{3x} dx \right)$$

$$\therefore f(x) = -\frac{8}{27} e^{-3x} + e^{-x} - \frac{x^2}{6} + \frac{1}{9} x + \frac{8}{27}$$

由题设可知 $a > 0$,

$$I = \int \sqrt{x^2 - a^2} dx = \frac{a}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln(x + \sqrt{x^2 - a^2}) + C, \quad (1)$$

下证 3. 若 $a > 0$, 则 $x > a$, 作代换

$$x = a \sec t = \frac{a}{\cos t}, \quad t \in (0, \frac{\pi}{2})$$

$$\text{则 } \cos t = \frac{a}{x}, \quad \frac{dx}{dt} = a \frac{\tan t}{\cos^2 t} > 0$$

$$\sqrt{x^2 - a^2} = \sqrt{a^2 \sec^2 t - a^2} = a \tan t$$

$$\therefore I = \int a \tan t \cdot a \frac{\tan t}{\cos^2 t} dt = a^2 \int \frac{\sin^2 t}{\cos^3 t} dt = a^2 \int \frac{1 - \cos^2 t}{\cos^3 t} dt$$

$$= a^2 \left(\int \frac{1}{\cos^3 t} dt - \int \sec t dt \right) \quad (2)$$

$$\text{记 } \frac{1}{\cos^3 t} = \frac{1}{\cos t} \cdot \frac{1}{\cos^2 t} = \frac{1}{\cos t} \cdot \frac{1}{1 - \sin^2 t} = \frac{1}{2} \frac{\sin t}{\cos^3 t} \quad (3)$$

$$\therefore \left(\frac{\sin t}{\cos^3 t} \right)' = \frac{\cos t \cdot \cos^3 t - \sin t \cdot 2 \cos t (-\sin t)}{\cos^6 t}$$

$$= \frac{\cos^4 t + 2 \sin^2 t \cos t}{\cos^6 t} = \frac{1 + \sin^2 t}{\cos^3 t} = \frac{1}{\cos^3 t} + \frac{\sin^2 t}{\cos^3 t}$$

$$\text{上式 (3) 代入 (2) 得}$$

$$\int \left(\frac{\sin t}{\cos^3 t} \right)' dt = \int \frac{1}{\cos^3 t} dt + \int \frac{1 - \cos^2 t}{\cos^3 t} dt = 2 \int \frac{1}{\cos^3 t} dt - \int \sec t dt \quad (4)$$

$$\Rightarrow \int \frac{1}{\cos^3 t} dt = \frac{1}{2} \left(\frac{\sin t}{\cos t} + \int \sec t dt \right) \quad (5)$$

(5) 代入 (2) 得

$$I = a^2 \left(\frac{1}{2} \frac{\sin t}{\cos^2 t} - \frac{1}{2} \int \sec t dt \right)$$

$$= \frac{1}{2} \frac{a}{\cos t} \cdot \frac{\sin t}{\cos t} - \frac{a^2}{2} \int \sec t dt$$

$$= \frac{1}{2} \frac{a \cdot \sin t}{\cos^2 t} - \frac{a^2}{2} \ln(x + \sqrt{x^2 - a^2}) + C \quad \text{⑥}$$

(由 P19 得)

$$\int \sec t dt = \ln |\sec t + \tan t| + C_1$$

$$= \ln \left(\frac{x}{a} + \frac{\sqrt{x^2 - a^2}}{a} \right) + C_1$$

$$= \ln(x + \sqrt{x^2 - a^2}) + C_1 - \ln a,$$

$$\therefore -\frac{a^2}{2} (C_1 - \ln a) \triangleq C_2 \quad \text{⑥ 得证}$$